

Industry trends

We looked at the recent launches in the world of solid-state memories and discovered the following broad trends:

More storage in lesser space. This trend is typically common across most components. In the flash memory space, companies like Toshiba have launched a 3D-stacked cell structure memory (BiCS 3D Memory) and claim that it features 3-bit-per-cell technology. Such chips are aimed at ultra-high density storage devices, which end up in applications like smartphones, tablets, memory cards, etc.

Interestingly, the new 3D NAND technology mutually developed by Micron and Intel also lays claim to be the world's highest density flash memory. Based on the floating gate cell technique, the innovative 3D NAND technology stacks flash cells vertically in 32 layers to achieve 256Gb multi-level cell (MLC) and 384Gb triple-level cell (TLC) dies that fit within a standard package. These capacities can enable gum-stick-sized SSDs with more than 3.5TB of storage and standard 6.35cm (2.5-inch) SSDs greater than 10TB!

Faster processing. Again, a no-brainer as speed is a key factor that affects selection of memory chips. But there is an entirely new category of memory competing with flash and gunning for the top spot. Recently, Imec and Panasonic have jointly developed an innovative non-volatile memory based on resistive RAM (ReRAM) technology, which can be embedded on to any 28nm processor to provide high-speed memory access.

Tighter security. Security in memory solutions is of paramount importance for highly scrutinised sectors including medical, banking and government industries. Micron has laid claim to the first TCG Enterprise-level hardware encryption for secured applications. Micron's M510DC SATA SSD, claims the company, is ideal for improving application load times for content delivery networks and enhancing response times to enable quicker decisions in big data analytics.

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accommodate more than 10TB. Since the growth in capacity is achieved by stacking the cells and not by bringing down their dimensions, both performance and endurance are increased, making even the TLC designs suitable for data centre storage.

Another innovation that helps increase the performance and reliability of this 3D NAND flash memory is the use of a floating gate cell design. In this architecture, the cells' transistors have a second insulated gate that retains the electrons till a strong external voltage is applied. This is a time-tested method frequently used in 2D NAND flash, which Intel has extended to their 3D design, too. Other features like sleep mode are also incorporated in the design to enable power savings, too.

SanDisk, in association with Toshiba, has also developed a 256-gigabit 3D NAND chip, with three bits per cell and 48 layers.

Samsung's three-dimensional vertical-NAND (3D V-NAND) technology has enabled it to make a 6.35cm SSD of around 16TB capacity. V-NAND involves both structural and material innovations. A new technology called the Channel Hole Technology enables the stacked cells to connect vertically through a cylindrical channel that runs

through these. This is supplemented with a material innovation called the Charge Trap Flash technology, which prevents data corruption caused by cell-to-cell interference. This technology uses a non-conductive layer of silicon nitride, which wraps around the control gate of a cell, acting as an insulator that temporarily traps electrical charges and maintains cell integrity. While current products and prototypes have 48 layers, Samsung says its technology is capable of 100 layers, which it hopes to achieve by 2017.

In March this year, Samsung officially announced the launch of PM1633a, a 15.36TB drive, for enterprise applications. The SSD is based on a 12-gigabit-per-second serial attached SCSI interface and comes in a 6.35cm form factor.

Keeping up with the networks

The latest networking standards seem to promise nothing short of lightning speed. The emerging Wi-Fi standard 802.11ac operates at 1300Mbps, Wi-Gig (802.11ad) that was introduced in Intel's sixth-generation Core PC platform can operate at 7000Mbps, universal serial bus (USB) 3.1 at 10,000Mbps and Thunderbolt 3 at 40,000Mbps.

Hard disk drives (HDDs) are hopelessly slow in comparison, and that is why no one imagines future personal computers to have these.

SSDs fare much better than HDDs, but even the current-generation serial AT attachment (SATA) SSDs—that typically have speeds of about 4400Mbps—do not seem any match for the latest networks. What is to be noted here is that, the NAND flash memory is not the bottleneck but the SATA interface is. In order to overcome this, the industry has standardised on a protocol called non-volatile memory express (NVMe) that communicates over the PCIe host interface. NVMe is considered to bode well for the SSD industry because this technology has been developed for solid-state storage, which means a lot of parallelism and other exceptional features have been designed into it. Plus, standardisation of this technology will eventually drive down the cost, making SSDs more affordable to everybody.

The NVMe interface scales up by using multiple lanes. For example, an NVMe SSD with four lanes has a speed of 32,000Mbps, which is many times faster than SATA SSDs. While SATA supports a single queue of up to 32 commands, NVMe supports 64,000 queues that can each hold up to 64,000 commands.

Let us say, you have a quad-core computer. NVMe will enable commands to be completed in parallel across all the four cores, while in the case of SATA the commands have to go through a single core. That way, NVMe increases throughput and reduces latency. With an NVMe SSD and a fast network connection, data sharing from your laptop will be super-fast. Experts feel that the speed of NVMe SSDs will also extend a laptop's battery life. If you take the example of a data transfer job that requires an HDD to operate at an active power of 6W for 180 seconds, an NVMe SSD can complete the same task in 10 seconds. This means 6W of active power for just 10 seconds, after